Shivam Juyal

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Summary

Software engineer building **LLM agents**, **RAG pipelines**, and **API integrations** in **TypeScript** and **Python**, with hands-on experience **debugging** production issues and documenting fixes. Fast learner, keen to work directly with customers.

Experience

Webrex Studio Jan 2026 – Mar 2026
MERN Developer Intern *Ahmedabad, Remote*

- Built the **graph-generation** step of a **Claude Code slash command** that maps a codebase into a **graph** (files as nodes with short summaries, linked by edges), and **integrated** the team's existing **hybrid semantic + keyword RAG pipeline** to find relevant nodes so Claude reads only those files and their neighbors instead of the whole repo.
- Enriched user prompts with graph context in a single pass, cutting back-and-forth with Claude; enrichment completes in **under 15 seconds** on a **500-file** repo.
- Built an **incremental, event-driven daemon** to replace a full graph rebuild on every significant change; it re-processes only changed files, cutting rebuild compute and LLM token usage by **roughly 75–80%** (typical changes touched 20–25% of files).
- Debugged production issues across **REST APIs**, **Node.js** services, and **MongoDB** by tracing logs to root cause.
- Collaborated within a 7-engineer **Agile** team via code reviews and daily standups, while documenting fixes and technical decisions.

Projects

Quill | *Next.js, Node.js, PostgreSQL, Redis, BullMQ, Python, Groq AI, Cohere Reranker* GitHub | Live Demo

- Built a RAG PDF Q&A app with **async BullMQ/Redis ingestion**, **pgvector** retrieval, and **Groq-to-Gemini failover** for rate limits and outages; added **GitHub Actions CI** running server tests on every push and pull request.
- Ran a retrieval experiment on a 50-question eval set: hybrid vector + keyword **RRF** with **Cohere re-ranking** raised **hit@1 from 54% (vector-only) to 80%** and **MRR@8 from 0.687 to 0.857**.
- Scored answers with LLM-as-Judge: **86.7% correctness, 92.7% faithfulness, 87.7% relevancy**.

Inflow | *TypeScript, Node.js, React, LangGraph, Groq AI, Gmail/Calendar API, Supabase, Zod, Vitest* GitHub

- Built an AI **email-automation** agent with **LangGraph** that classifies Gmail into 6 categories at **88%** accuracy on **200** emails, using Zod-validated LLM output and OAuth 2.0.
- Designed a **human-in-the-loop** flow (drafts and meetings only after approval), with **idempotent processing** to prevent duplicate replies; wrote **175 automated Vitest tests** to verify it.

PushCloud | *React, Node.js, Docker, AWS (ECS Fargate, S3), Redis, Socket.io, MinIO, Supabase, Vitest* GitHub | Demo Video

- Built a multi-tenant platform serving React/Vite apps via **reverse-proxy** routing; migrated from **AWS (ECS Fargate, S3)** to **Docker** for lower cost and control; load-tested **10 concurrent builds**, passing **18/18 checks**.
- Added tenant-isolated **Socket.io** build-log streaming; wrote **Vitest** security tests that blocked **39/39 unauthorized attempts and 44/44 injection payloads**, with no log loss or reordering across **700 logs**.

Technical Skills

Languages & Frontend: TypeScript, JavaScript, Python, SQL, React, Next.js
AI & LLM: AI Agents, RAG, LangGraph, LLM APIs (Groq, Gemini), Workflow Automation, pgvector
Backend & APIs: Node.js, Express.js, REST APIs, API Integrations, OAuth 2.0, BullMQ
Databases: PostgreSQL, MongoDB, Redis, Supabase
Testing & DevOps: Vitest, Postman, Docker, AWS (ECS Fargate, S3), Linux, GitHub Actions (CI/CD), Git

Education

Graphic Era Hill University Dehradun, Uttarakhand
Bachelor of Technology in Computer Science and Engineering (7.22/10 CGPA) *June 2021 – July 2025*

Achievements

Built and published **@the-white-wolf/ai-commit** to npm, a CLI that generates commit messages from diffs, reaching **1800+** downloads.
Solved **1,000+** Data Structures and Algorithms problems across competitive programming platforms; reached a peak **LeetCode** rating of **1,700+**.